

**B.E. Electrical Engineering
Second Year Second Semester
Power Supply Systems
Part-II Complete Solutions
2025 Question Paper**

Solved in Simple SN Ma'am Style

This document follows the method:

Definition → Diagram → Formula
Stepwise solution → Final answer

All diagrams are kept simple, labelled, and margin-safe.

Prepared for semester examination revision

May 15, 2026

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How to Use This Solution Sheet

SN Ma'am Method

For every answer, write in this order in the exam:

1. Start with a very small definition.
2. Draw a neat labelled diagram if the topic is diagram-based.
3. Write assumptions before derivation or numerical solution.
4. Use KCL, KVL, voltage drop, current loading, or annual cost equation as required.
5. Box the final answer with proper units.

In this paper, Part-II mainly contains: distributor design, radial/ring distribution, Kelvin's law, conductor comparison, three-wire DC system, substation, HRC fuse, neutral grounding, earthing, and high tension service connection.

1. Question 1: DC Distributor with Uniform and Concentrated Loading

Question

A DC distributor shown in Fig. Q1 has a uniform load of 1 A/m in addition to concentrated loads. The total resistance of the distributor is 0.075Ω , and it is fed at 230 V at A. Determine the voltages at each load point. Derive the necessary formula.

1.1. Given Data

Supply voltage at A	$V_A = 230 \text{ V}$
Uniform load	$i = 1 \text{ A/m}$
Total length	$l = 50 + 100 + 100 + 150 + 100 = 500 \text{ m}$
Total distributor resistance	$R = 0.075 \Omega$
Resistance per metre	$r = R/l = 0.075/500 = 0.00015 \Omega/\text{m}$
Concentrated loads	B: 100 A, C: 50 A, D: 50 A, E: 100 A

1.2. Neat Diagram of the Given Distributor

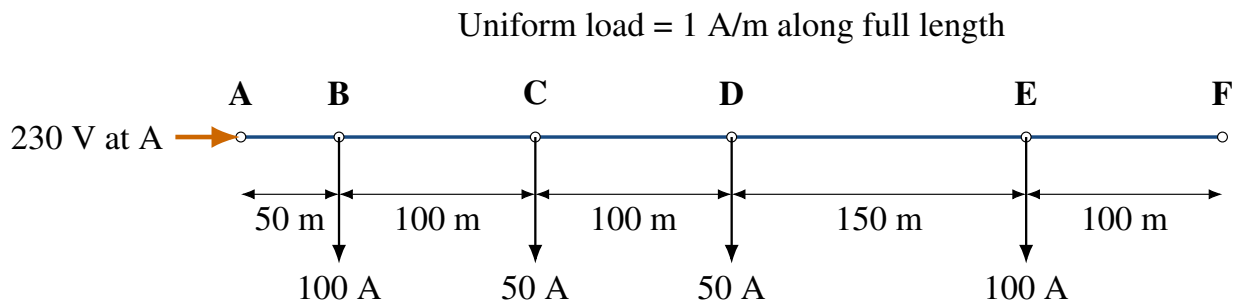


Figure 1: DC distributor fed at one end with uniform and concentrated loads.

1.3. Derivation of Necessary Formula

Definition / Main idea

For a distributor fed at one end, the current in a section is equal to the sum of all loads beyond that section. Therefore voltage drop in a section is:

$$\text{Voltage drop} = \text{section current} \times \text{section resistance.}$$

Let the total length of the distributor be L . Let a point be situated at a distance x from feeding end A. Let the uniform load be i A/m and resistance per metre be $r\Omega/\text{m}$.

At a distance s from A, the uniform load to the right is:

$$i(L-s)$$

So drop due to uniform loading from A to distance x is:

$$\Delta V_u = \int_0^x i(L-s)r \, ds$$

$$\Delta V_u = ir \left(Lx - \frac{x^2}{2} \right)$$

If a concentrated load I_k is at distance x_k , then its contribution to drop up to point x is:

$$\Delta V_k = rI_k \min(x, x_k)$$

Therefore total drop up to point x is:

$$\Delta V_x = ir \left(Lx - \frac{x^2}{2} \right) + r \sum I_k \min(x, x_k)$$

SN Ma'am Method

Use this formula directly for B, C, D, E and F. Here, $L = 500\text{m}$, $i = 1\text{ A/m}$, and $r = 0.00015\Omega/\text{m}$.

1.4. Stepwise Calculation

Positions of load points from A are:

$$B = 50\text{m}, \quad C = 150\text{m}, \quad D = 250\text{m}, \quad E = 400\text{m}, \quad F = 500\text{m}$$

At B, $x = 50\text{m}$

$$\Delta V_B = 0.00015 \left[1 \left(500 \times 50 - \frac{50^2}{2} \right) + 100(50) + 50(50) + 50(50) + 100(50) \right]$$

$$\Delta V_B = 5.8125\text{ V}$$

$$V_B = 230 - 5.8125 = 224.1875\text{ V}$$

At C, $x = 150\text{m}$

$$\Delta V_C = 14.8125\text{ V}$$

$$V_C = 230 - 14.8125 = 215.1875\text{ V}$$

At D, $x = 250\text{m}$

$$\Delta V_D = 21.5625\text{ V}$$

$$V_D = 230 - 21.5625 = 208.4375\text{ V}$$

At E, $x = 400\text{m}$

$$\Delta V_E = 27.75\text{ V}$$

$$V_E = 230 - 27.75 = 202.25\text{ V}$$

At F, $x = 500\text{m}$

$$\Delta V_F = 28.50\text{ V}$$

$$V_F = 230 - 28.50 = 201.50\text{ V}$$

1.5. Answer Table

Point	Distance from A (m)	Voltage drop from A (V)	Voltage at point (V)
B	50	5.8125	224.1875
C	150	14.8125	215.1875
D	250	21.5625	208.4375
E	400	27.7500	202.2500
F	500	28.5000	201.5000

Final Answer

$$V_B = 224.19\text{ V}, \quad V_C = 215.19\text{ V}, \quad V_D = 208.44\text{ V}$$

$$V_E = 202.25\text{ V}, \quad V_F = 201.50\text{ V}$$

The minimum voltage occurs at the farthest end F.

2. Question 2: Feeders, Distributors, Radial and Ring Main Systems

Question 2(i)

Explain the basis for the design of distributors and feeders in the power supply system.

Definition / Main idea

A feeder carries electrical power from a substation to the feeding point of a distributor. A distributor is a conductor from which tapings are taken for supplying consumers.

2.1. Basic Supply Layout

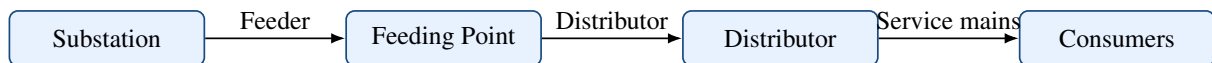


Figure 2: Feeder, distributor and service mains.

2.2. Basis of Feeder Design

A feeder has no intermediate tapings normally. It only carries current from substation to distributor.

Therefore the main design basis of feeder is:

Current carrying capacity

The voltage drop in the feeder is not the primary design factor because no consumer is directly tapped along the feeder.

2.3. Basis of Distributor Design

A distributor has many tapings. Consumers are connected at different points. So voltage at all consumer terminals should remain within permissible limits.

Therefore the main design basis of distributor is:

Voltage drop limitation

2.4. Comparison Table

Point	Feeder	Distributor
Function	Carries power from substation to distributor	Supplies power to consumers through tapplings
Tappings	Normally no tapping	Many tapplings are taken
Design basis	Current carrying capacity	Permissible voltage drop
Voltage drop	Less important	Very important
Fault effect	Affects connected distributor	Affects consumer supply directly

Final Answer

Feeder is designed mainly from current carrying capacity. Distributor is designed mainly from voltage drop consideration.

Question 2(ii)

Differentiate between radial system, ring main system and ring main with interconnector for a distribution system.

2.5. Radial System

Definition / Main idea

In radial distribution system, each distributor is fed from one end only.

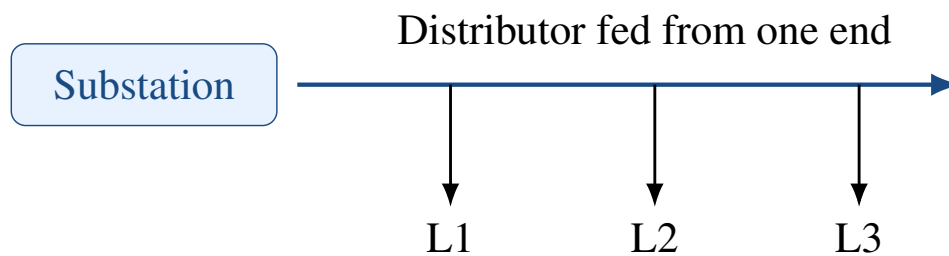


Figure 3: Radial distribution system.

Features:

- Simple and cheap.
- Voltage drop is maximum at far end.
- Reliability is poor because fault near source interrupts all consumers beyond it.

2.6. Ring Main System

Definition / Main idea

In ring main system, the distributor forms a closed loop and is fed from one or more points.

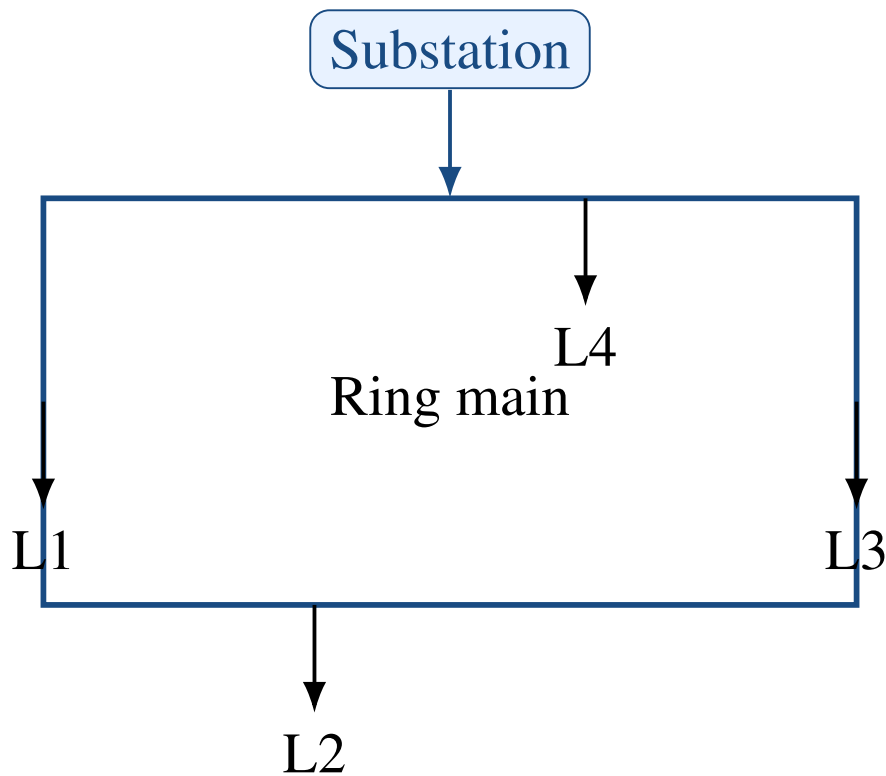


Figure 4: Ring main distribution system.

Features:

- Better reliability than radial system.
- Supply can reach a point from two directions.
- Copper saving is possible with proper feeding points.

2.7. Ring Main with Interconnector

Definition / Main idea

A ring main with interconnector has an additional connection between two points of the ring. This improves voltage profile and reliability.

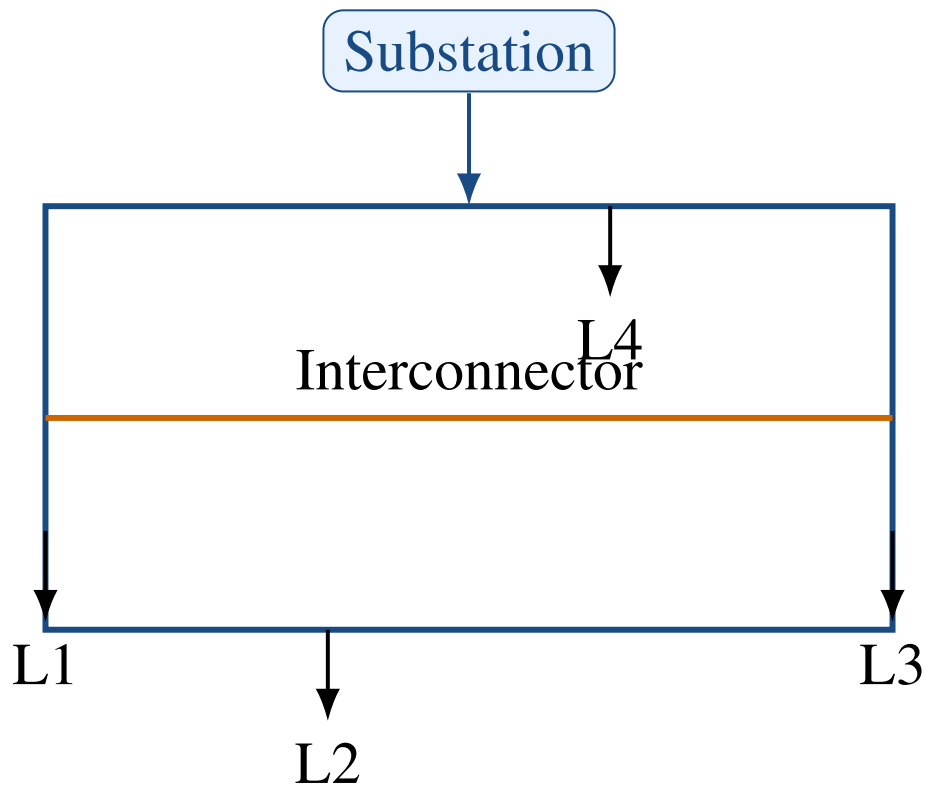


Figure 5: Ring main with interconnector.

2.8. Difference Table

Point	Radial system	Ring main	Ring main with inter-connector
Connection	Fed from one end	Closed loop	Closed loop with extra link
Cost	Lowest	Higher than radial	Highest among three
Reliability	Poor	Good	Very good
Voltage regulation	Poor at far end	Better	Best
Fault effect	Large interruption	Faulty section can be isolated	More flexible isolation possible
Use	Small loads and rural areas	Urban distribution	Important urban/industrial area

Final Answer

Radial system is cheapest but least reliable. Ring main gives better continuity of supply. Ring main with interconnector gives the best voltage regulation and reliability, but cost is higher.

3. Question 3: Kelvin's Law and Economical Current Density

Question 3(i)

What are the limitations of Kelvin's law?

3.1. Limitations

1. It is difficult to estimate energy loss accurately without knowing the exact load curve.
2. The assumption that capital charge varies linearly with conductor area is not strictly true.
3. It does not consider mechanical strength, corona loss, safe current density and voltage drop limits.
4. The economical area obtained may be too small for mechanical strength.
5. Interest and depreciation charges cannot be predicted exactly.

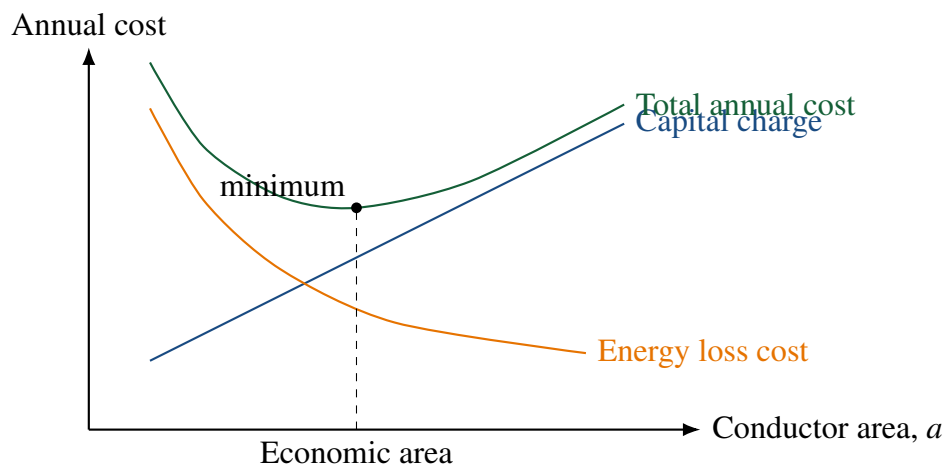


Figure 6: Kelvin's law cost curve.

Question 3(ii)

Determine the most economical current density in a three-phase 11 kV cable which supplies 1500 kW at 0.8 p.f. lagging for 300 days in a year at an average rate of 8 hours per day. The resistance per km length of cable of 1 sq. cm cross-section is 0.173Ω . The rate of interest and depreciation is 12%. Cost of energy loss is 2 paisa per unit and capital cost per km of cable is Rs. $(8000 + 20000a)$, where a is cross-sectional area in sq. cm.

3.2. Theory Used

Definition / Main idea

Kelvin's law states that the most economical conductor size is obtained when the variable annual charge on capital cost is equal to the annual cost of energy lost in the conductor.

$$\text{Variable annual capital charge} = \text{Annual cost of energy loss}$$

3.3. Given Data

Power supplied	$P = 1500 \text{ kW}$
Line voltage	$V_L = 11 \text{ kV}$
Power factor	$\cos \phi = 0.8$
Operating time per year	$300 \times 8 = 2400 \text{ h}$
Resistance per km for 1 sq. cm	0.173Ω
Interest and depreciation	12%
Cost of energy loss	2 paisa/unit = Rs. 0.02/kWh
Capital cost per km	Rs. $(8000 + 20000a)$

3.4. Step 1: Line Current

For 3-phase system:

$$P = \sqrt{3} V_L I \cos \phi$$

$$I = \frac{P}{\sqrt{3} V_L \cos \phi}$$

$$I = \frac{1500 \times 10^3}{\sqrt{3} \times 11000 \times 0.8}$$

$$I = 98.41 \text{ A}$$

3.5. Step 2: Resistance Per km Per Conductor

If conductor area is a sq. cm, then

$$R = \frac{0.173}{a} \Omega/\text{km}$$

3.6. Step 3: Annual Energy Loss

Copper loss for 3 conductors:

$$P_{\text{loss}} = 3I^2R = 3I^2 \frac{0.173}{a} \text{ W per km}$$

Energy lost per year:

$$E = \frac{3I^2(0.173) \times 2400}{1000a} \text{ kWh per km}$$

Cost of energy lost:

$$C_e = \frac{3I^2(0.173) \times 2400}{1000a} \times 0.02$$

Substituting $I = 98.41$ A:

$$C_e = \frac{241.27}{a} \text{ Rs per year per km}$$

3.7. Step 4: Variable Annual Capital Charge

Variable part of capital cost:

$$20000a$$

Annual charge at 12%:

$$C_c = 0.12 \times 20000a = 2400a$$

3.8. Step 5: Apply Kelvin's Law

$$2400a = \frac{241.27}{a}$$

$$a^2 = \frac{241.27}{2400} = 0.10053$$

$$a = 0.317 \text{ sq. cm}$$

3.9. Step 6: Economical Current Density

$$\text{Current density} = \frac{I}{a}$$

$$J = \frac{98.41}{0.317} = 310.4 \text{ A/sq. cm}$$

Since 1 sq. cm = 100sq. mm,

$$J = 3.10 \text{ A/sq. mm}$$

Final Answer

Most economical area = 0.317sq. cm

Most economical current density = 310.4 A/sq. cm = 3.10 A/sq. mm

4. Question 4: Conductor Material Comparison

Question 4(i)

Explain why the maximum voltage between each conductor and earth forms the basis for comparison of volume of conductor for overhead transmission line and for underground cables maximum voltage between two conductors is considered.

4.1. Simple Explanation

Definition / Main idea

The basis of comparison is chosen according to the main insulation stress in the system.

4.2. Overhead Transmission Line

In overhead lines, conductors are supported on poles or towers by insulators. Each conductor is insulated from earth through the insulator string and air.

Therefore the important insulation stress is between:

conductor and earth

So for overhead transmission, comparison is based on maximum voltage between each conductor and earth.

4.3. Underground Cable

In underground cables, conductors are very close to each other and are surrounded by insulation. The maximum dielectric stress appears between conductors.

Therefore the important insulation stress is between:

conductor and conductor

So for underground cable systems, comparison is based on maximum voltage between two conductors.

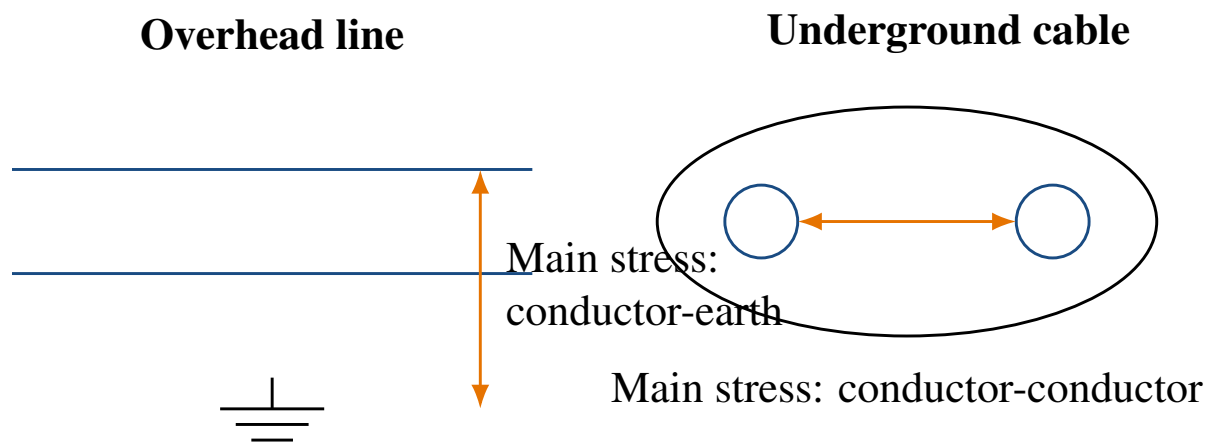


Figure 7: Basis of comparison for overhead and underground systems.

Final Answer

For overhead lines, the conductor is mainly insulated from earth, so conductor-earth voltage is considered. For underground cables, conductors are close together inside insulation, so conductor-conductor voltage is considered.

Question 4(ii)

Assuming transmitted power, maximum voltage between conductors, length of transmission line, power factor and transmission efficiency remaining same for all cases, compare the volume of copper required in an underground scheme for (i) AC single phase 3-wire system and (ii) AC three phase 3-wire system.

4.4. Common Assumptions

1. Same power P is transmitted.
2. Same length l is used.
3. Same line loss P_{loss} is allowed.
4. Same voltage between conductors V is used.
5. Same power factor $\cos \phi$ is used.
6. Resistance of one conductor is $R = \rho l / a$.

4.5. Case 1: AC Single Phase 3-Wire System

For balanced loading, neutral current is zero. But the neutral conductor is still provided. In ordinary calculation, neutral cross-section is taken as half of each outer conductor.

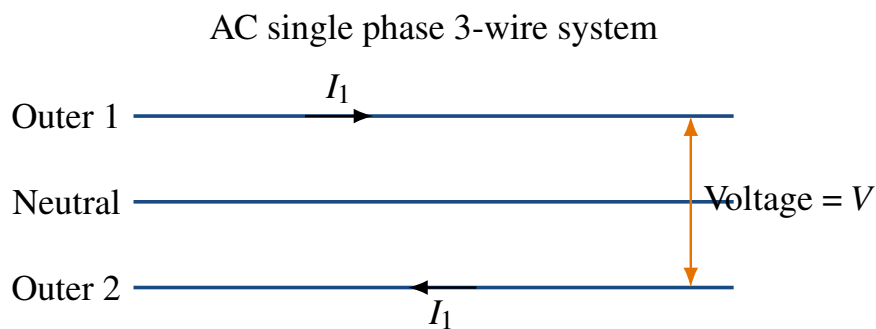


Figure 8: Single phase 3-wire underground scheme.

Power transmitted:

$$P = VI_1 \cos \phi$$

$$I_1 = \frac{P}{V \cos \phi}$$

Line loss in two outer conductors:

$$P_{\text{loss}} = 2I_1^2 \frac{\rho l}{a_1}$$

$$P_{\text{loss}} = 2 \left(\frac{P}{V \cos \phi} \right)^2 \frac{\rho l}{a_1}$$

$$a_1 = \frac{2P^2 \rho l}{P_{\text{loss}} V^2 \cos^2 \phi}$$

Volume of copper:

$$V_1 = \left(2a_1 + \frac{a_1}{2} \right) l = 2.5a_1 l$$

$$V_1 = \frac{5P^2 \rho l^2}{P_{\text{loss}} V^2 \cos^2 \phi}$$

4.6. Case 2: AC Three Phase 3-Wire System

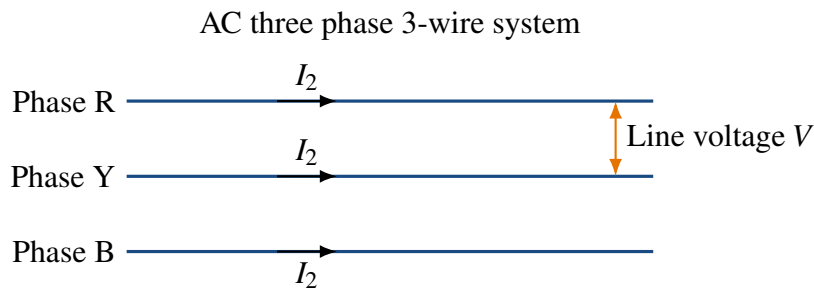


Figure 9: Three phase 3-wire underground scheme.

Power transmitted:

$$P = \sqrt{3} V I_2 \cos \phi$$

$$I_2 = \frac{P}{\sqrt{3} V \cos \phi}$$

Line loss in three conductors:

$$P_{\text{loss}} = 3 I_2^2 \frac{\rho l}{a_2}$$

$$P_{\text{loss}} = 3 \left(\frac{P}{\sqrt{3} V \cos \phi} \right)^2 \frac{\rho l}{a_2}$$

$$a_2 = \frac{P^2 \rho l}{P_{\text{loss}} V^2 \cos^2 \phi}$$

Volume of copper:

$$V_2 = 3a_2 l$$

$$V_2 = \frac{3P^2 \rho l^2}{P_{\text{loss}} V^2 \cos^2 \phi}$$

4.7. Comparison

$$\frac{V_1}{V_2} = \frac{5P^2\rho l^2}{3P^2\rho l^2}$$

$$\boxed{\frac{V_1}{V_2} = \frac{5}{3}}$$

Final Answer

Copper required by single phase 3-wire : copper required by 3-phase 3-wire = 5 : 3

So, AC single phase 3-wire system requires $\frac{5}{3}$ times the copper required by AC three phase 3-wire system.

5. Question 5: Three-Wire DC Distributor and Effect of Transmission Voltage

Question 5(i)

A three-wire DC distributor 600 m long is supplied at 500/250 V and is loaded as follows:

Load between positive terminal and neutral wire: 60 A, 200 m from positive terminal; 40 A, 360 m from positive terminal.

Load between neutral wire and negative terminal: 20 A, 100 m from negative terminal; 60 A, 260 m from negative terminal; and 15 A, 600 m from negative terminal.

The resistance of each outer is 0.02Ω per 100 m, and the cross-section of the neutral wire is same as the outer. Find the voltage across each load point.

5.1. Neat Diagram

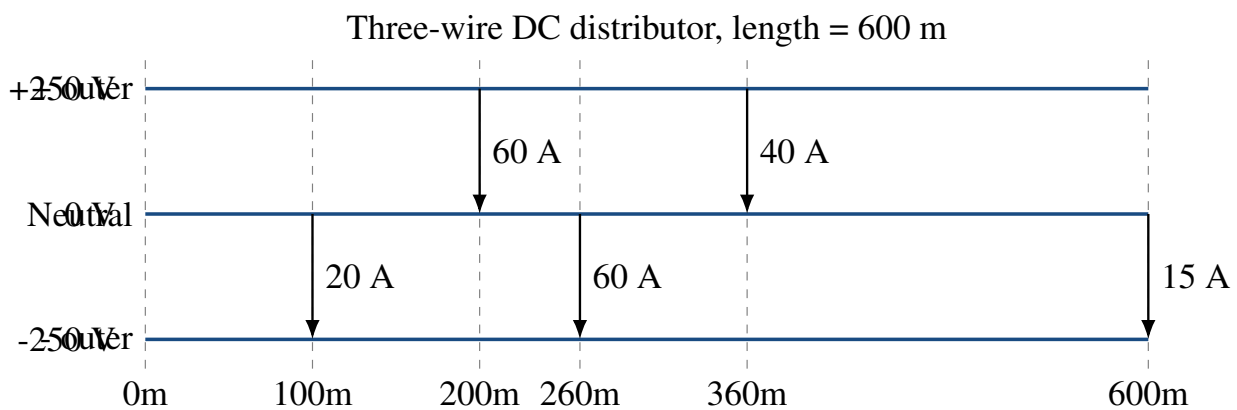


Figure 10: Three-wire DC distributor with positive-neutral and neutral-negative loads.

5.2. Given Data

$$R_{\text{per m}} = \frac{0.02}{100} = 0.0002 \Omega/\text{m}$$

Supply voltages at feeding end:

$$V_{PN} = 250 \text{ V}, \quad V_{NM} = 250 \text{ V}, \quad V_{PM} = 500 \text{ V}$$

5.3. Current Distribution

Section	Length (m)	Positive outer current (A)	Negative outer current (A)
0-100	100	100	95
100-200	100	100	75
200-260	60	40	75
260-360	100	40	15
360-600	240	0	15

Neutral current in each section is the algebraic difference of positive-side and negative-side requirements. It is found by KCL at each tapping point.

Section	Neutral current (A)	Direction meaning
0-100	-5	Towards feeding end
100-200	-25	Towards feeding end
200-260	35	Away from feeding end
260-360	-25	Towards feeding end
360-600	15	Away from feeding end

5.4. Calculated Potentials

Let neutral at feeding end be 0 V. Then:

$$V_+(0) = 250 \text{ V}, \quad V_N(0) = 0 \text{ V}, \quad V_-(0) = -250 \text{ V}$$

The conductor drop in each section is:

$$\Delta V = I \times r \times l$$

The resulting potentials at important points are shown below.

Distance (m)	V_+ (V)	V_N (V)	V_- (V)	Remark
100	248.00	0.10	-248.10	Negative-side load point
200	246.00	0.60	-246.60	Positive-side load point
260	245.52	0.18	-245.70	Negative-side load point
360	244.72	0.68	-245.40	Positive-side load point
600	244.72	-0.04	-244.68	Negative-side load point

5.5. Voltage Across Each Load

For positive-neutral load:

$$V_{PN}(x) = V_+(x) - V_N(x)$$

For neutral-negative load:

$$V_{N-}(x) = V_N(x) - V_-(x)$$

Load type	Distance (m)	Load current (A)	Voltage across load (V)
Positive-neutral	200	60	$246.00 - 0.60 = 245.40$
Positive-neutral	360	40	$244.72 - 0.68 = 244.04$
Neutral-negative	100	20	$0.10 - (-248.10) = 248.20$
Neutral-negative	260	60	$0.18 - (-245.70) = 245.88$
Neutral-negative	600	15	$(-0.04) - (-244.68) = 244.64$

Final Answer

$$V_{PN,200m} = 245.40 \text{ V}, \quad V_{PN,360m} = 244.04 \text{ V}$$

$$V_{N-,100m} = 248.20 \text{ V}, \quad V_{N-,260m} = 245.88 \text{ V}, \quad V_{N-,600m} = 244.64 \text{ V}$$

Question 5(ii)

For transmission of electrical power through a transmission line, show that with increase in transmission line voltage, the volume of conductor material required decreases.

5.6. Derivation

Let power transmitted be P , voltage be V , power factor be $\cos \phi$, and current be I .

For a transmission line:

$$P = VI \cos \phi$$

$$I = \frac{P}{V \cos \phi}$$

Power loss in line:

$$P_{\text{loss}} = I^2 R$$

where

$$R = \frac{\rho l}{a}$$

So,

$$P_{\text{loss}} = \left(\frac{P}{V \cos \phi} \right)^2 \frac{\rho l}{a}$$

Rearranging,

$$a = \frac{P^2 \rho l}{P_{\text{loss}} V^2 \cos^2 \phi}$$

Volume of conductor:

$$\text{Volume} = al$$

$$\text{Volume} \propto \frac{1}{V^2}$$

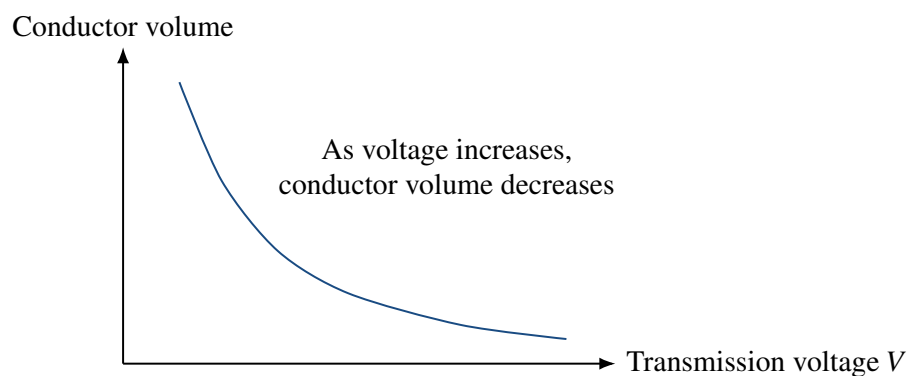


Figure 11: Conductor volume decreases with increase in transmission voltage.

Final Answer

Since conductor volume is inversely proportional to the square of voltage,

higher transmission voltage $\Rightarrow$ lower current $\Rightarrow$ lower conductor volume

6. Question 6: Substations, Isolator and Circuit Breaker

Question 6(i)

Discuss how the substation can be classified.

Definition / Main idea

A substation is an assembly of electrical apparatus used to change voltage level, improve power control, protect the system, and distribute power to consumers.

6.1. Classification Flowchart

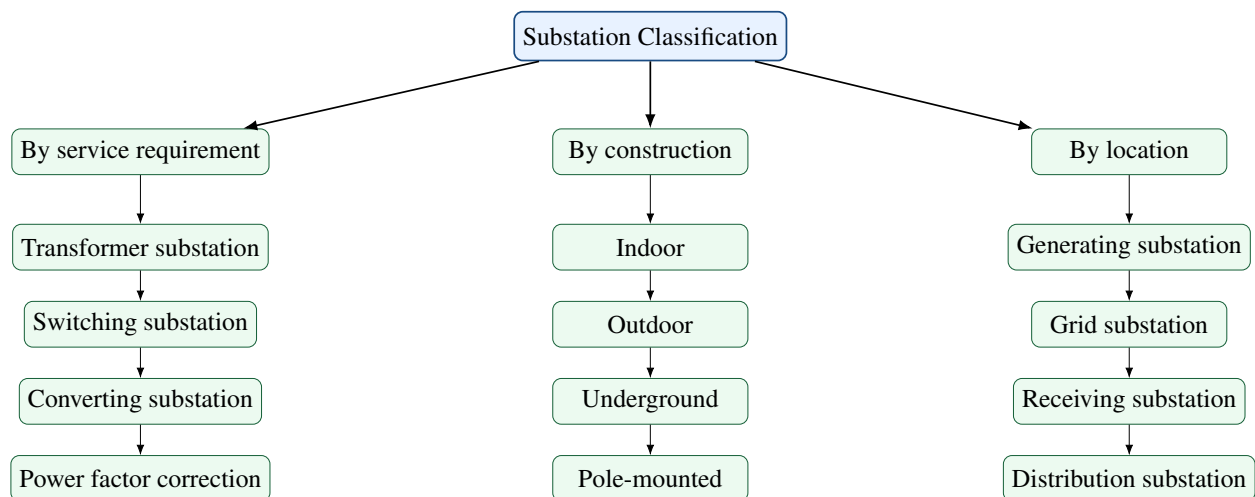


Figure 12: Classification of substations.

6.2. Classification in Simple Points

1. According to service requirement

- **Transformer substation:** changes voltage level by transformer.
- **Switching substation:** performs switching operation without changing voltage.
- **Converting substation:** converts AC to DC or DC to AC.
- **Power factor correction substation:** improves power factor by capacitors or synchronous condensers.

2. According to construction

- **Indoor substation:** equipment placed inside building, generally up to medium voltage.

- **Outdoor substation:** equipment placed in open air, used for high voltage.
- **Underground substation:** used where land is costly.
- **Pole-mounted substation:** small distribution transformer mounted on pole.

3. According to location and purpose

- Generating substation.
- Grid substation.
- Receiving substation.
- Distribution substation.

Final Answer

Substations are classified according to service requirement, construction, voltage level, and location/purpose. In exam, write one line use of each type.

Question 6(ii)

Mention the functions of isolator and circuit breaker in a substation.

6.3. Isolator

Definition / Main idea

An isolator is a mechanical switch used to isolate a part of the circuit from live supply under no-load condition.

Functions of isolator:

1. It disconnects a circuit part for maintenance.
2. It provides visible isolation for safety.
3. It is operated only when the circuit breaker is already open.
4. It does not interrupt fault current or load current.

6.4. Circuit Breaker

Definition / Main idea

A circuit breaker is a protective switching device which can make and break normal current as well as fault current.

Functions of circuit breaker:

1. It interrupts fault current.
2. It protects equipment from short circuit and overload.
3. It works with protective relays.
4. It can be used for normal switching operation also.

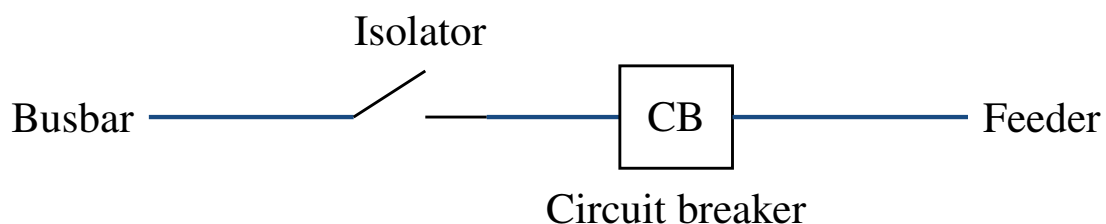


Figure 13: Position of isolator and circuit breaker in a feeder bay.

Final Answer

Isolator gives safe visible isolation but cannot break fault current. Circuit breaker breaks fault current and protects the system.

7. Question 7: HRC Fuse and Neutral Grounding

Question 7(i)

Discuss the principle of operation of the HRC fuse with a suitable diagram.

Definition / Main idea

HRC fuse means High Rupturing Capacity fuse. It is a fuse which can safely interrupt very high fault current without damage to surrounding equipment.

7.1. Construction

An HRC fuse generally consists of:

- heat-resisting ceramic cartridge,
- silver or copper fuse element,
- quartz/silica filling powder,
- end caps or contacts,
- sometimes a fuse indicator.

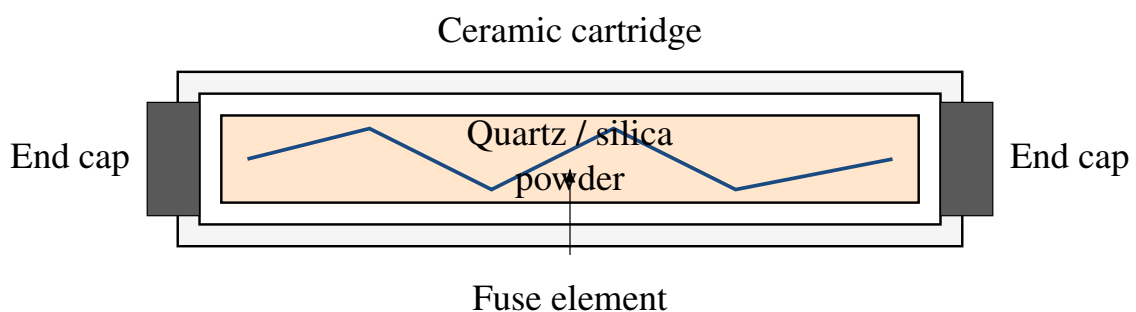


Figure 14: Construction of an HRC fuse.

7.2. Working Principle

1. Under normal current, the fuse element carries current without melting.
2. During fault, current becomes very high.
3. Heat produced is proportional to I^2R , so the fuse element melts quickly.
4. Arc is formed after melting.
5. Quartz powder absorbs heat and converts into a high-resistance substance.

6. Arc is quenched and the circuit is interrupted safely.

7.3. Time-current Characteristic

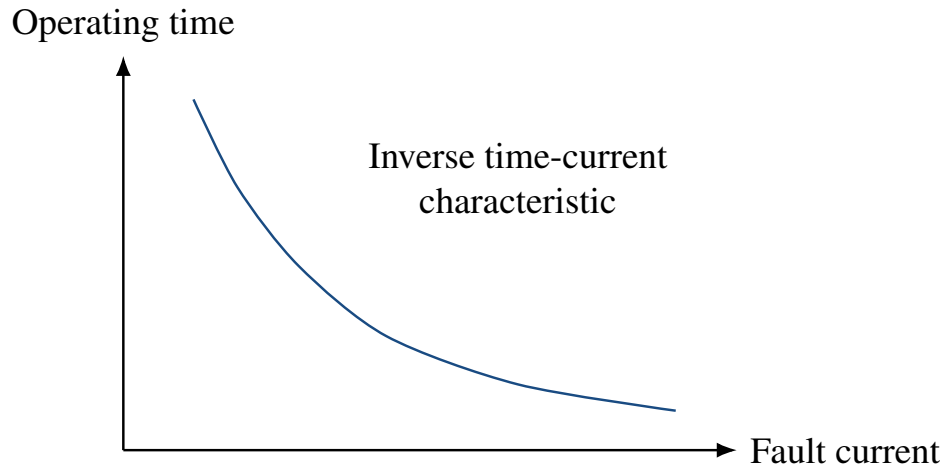


Figure 15: Fuse inverse time-current characteristic.

7.4. Advantages of HRC Fuse

- High breaking capacity.
- Fast operation under fault.
- No smoke or noise.
- Reliable and simple.
- Low maintenance.
- Good current limiting property.

Final Answer

An HRC fuse operates by melting of fuse element due to high I^2R heating. The quartz powder cools and quenches the arc, so the fault current is interrupted safely.

Question 7(ii)

Mention the advantages of neutral grounding.

Definition / Main idea

Neutral grounding means connecting the neutral point of a power system to earth either directly or through resistance, reactance or arc suppression coil.

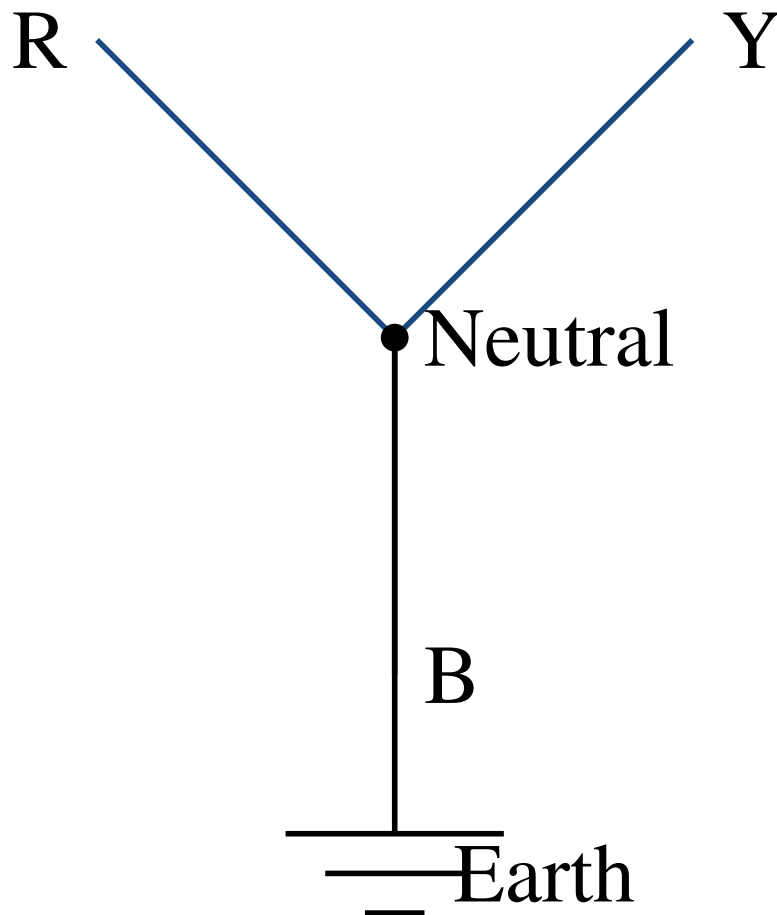


Figure 16: Neutral grounding of a three-phase system.

7.5. Advantages

1. It keeps the neutral point at earth potential.
2. It reduces overvoltage during line-to-ground fault.
3. It provides a path for earth fault current, so protective relays can operate.
4. It improves safety of equipment and persons.
5. It reduces arcing ground and insulation stress.

6. It improves system stability and reliability.

Final Answer

Neutral grounding makes earth fault detection easy, reduces dangerous overvoltages and protects equipment insulation.

8. Question 8: Pipe Earthing, Plate Earthing and HT Service Main

Question 8(i)

What do you understand by pipe and plate earthing?

Definition / Main idea

Earthing means connecting the metallic body of electrical equipment to the general mass of earth through a low-resistance conductor and earth electrode.

8.1. Need of Earthing

- To protect human beings from electric shock.
- To protect equipment from leakage current and fault current.
- To provide a low-resistance path for fault current.
- To help protective devices operate quickly.

8.2. Pipe Earthing

In pipe earthing, a galvanized iron pipe is buried vertically in the ground. Charcoal and salt are placed around the pipe to reduce earth resistance. Water is poured through a funnel to maintain moisture.

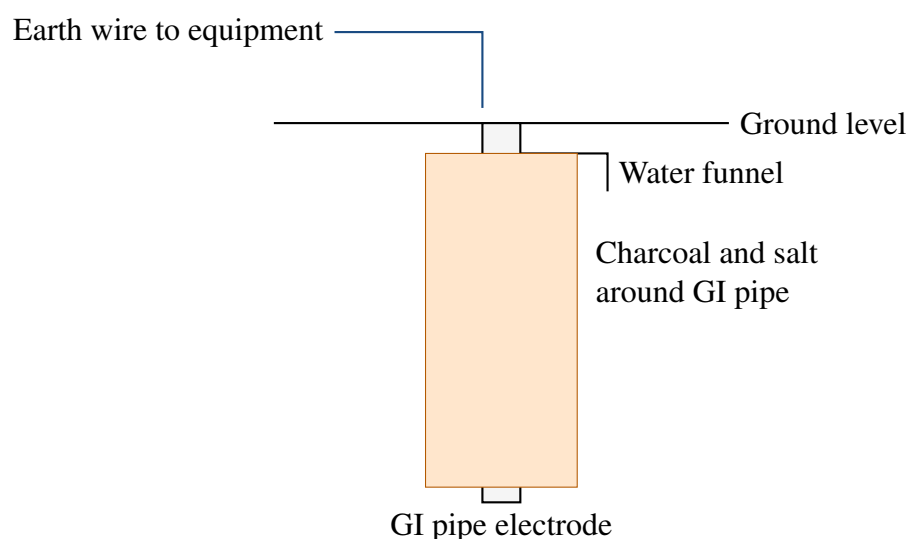


Figure 17: Pipe earthing.

8.3. Plate Earthing

In plate earthing, a copper or GI plate is buried vertically in the ground. The plate is surrounded by layers of charcoal and salt. Earth wire is bolted to the plate.

Usual plate sizes:

- Copper plate: 60 cm × 60 cm × 3.18 mm
- GI plate: 60 cm × 60 cm × 6.35 mm

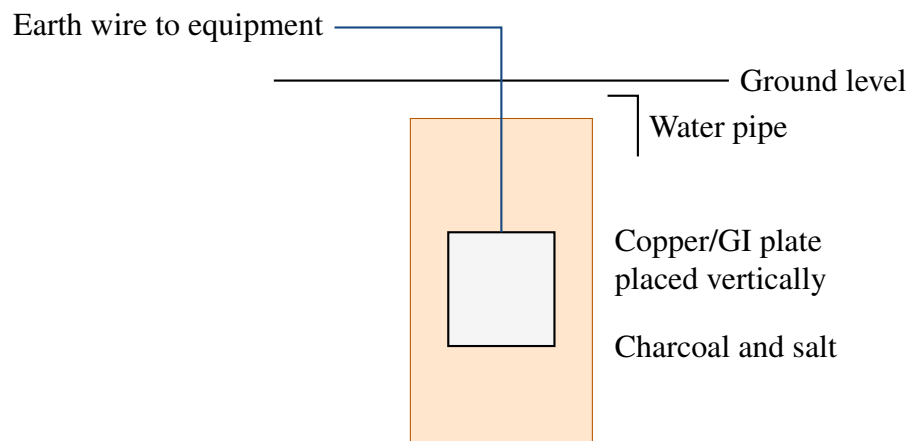


Figure 18: Plate earthing.

8.4. Pipe Earthing vs Plate Earthing

Point	Pipe earthing	Plate earthing
Electrode	GI pipe	Copper or GI plate
Cost	Cheaper	Costlier
Installation	Easier	Requires larger pit
Use	Common in domestic/industrial installation	Used where low earth resistance is needed
Maintenance	Watering needed	Watering needed

Final Answer

Pipe earthing uses a GI pipe as earth electrode. Plate earthing uses a copper or GI plate as earth electrode. In both methods charcoal, salt and moisture are used to reduce earth resistance.

Question 8(ii)

What are the advantages of high tension service main connections?

Definition / Main idea

High tension service main connection means supply is given to a consumer at high voltage, generally for large connected load. The consumer uses a transformer substation at his premises to step down the voltage.

8.5. Simple Layout

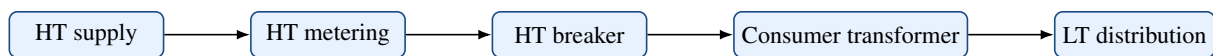


Figure 19: High tension service main connection.

8.6. Advantages

1. **Lower current:** For the same power, high voltage reduces current.

$$I = \frac{P}{\sqrt{3}V \cos \phi}$$

2. **Lower copper loss:** Since I is reduced, I^2R loss decreases.
3. **Better voltage regulation:** Voltage drop becomes smaller at higher voltage.
4. **Smaller conductor size:** Less current means smaller conductor size can be used.
5. **Suitable for large motors and industrial loads:** Heavy loads can be supplied without disturbing other consumers.
6. **Better protection:** Earth fault and short-circuit protection becomes more effective.
7. **Possible tariff advantage:** Large consumers may get better tariff in HT supply.

Final Answer

High tension service connection gives lower current, lower voltage drop, lower losses, better voltage regulation, better protection and suitability for heavy industrial loads.

Final Revision Table for Part-II 2025

Question	Core topic	Main exam weapon
Q1	DC distributor with uniform and concentrated load	Current loading, voltage drop formula, section-wise voltage
Q2	Feeders, distributors, radial and ring main	Definition + diagram + comparison table
Q3	Kelvin's law	Cost curve + annual cost equation + current density numerical
Q4	Conductor comparison	Assumptions + loss equation + volume ratio
Q5	Three-wire DC distributor	Neutral current + voltage across each load
Q6	Substation and switching devices	Classification flowchart + isolator vs circuit breaker
Q7	HRC fuse and neutral grounding	Construction diagram + working + advantages
Q8	Earthing and HT service main	Pipe/plate diagrams + HT supply advantages

SN Ma'am Style Final Checklist

SN Ma'am Method

Before writing any Part-II answer, check these points:

1. Did I write a simple definition?
2. Did I draw a clean labelled diagram?
3. Did I mention assumptions before derivation?
4. Did I use KCL/KVL/current-loading/voltage-drop clearly?
5. Did I show units in numericals?
6. Did I box the final answer?
7. Did I add one final exam remark?

Final Answer

For high marks, do not write only formulas. Show the examiner that you understand the physical system: where current flows, where voltage drops, and why the selected design is used.